Algorithms and changes in FOCEX 9.0–9.9

K. Esfarjani

November 2026

Abstract

FOCEX extracts force constants (FCs) of ranks 1 to 8 from force–displacement data of density-functional calculations. Between 23 October and 6 November 2026 the code went from version 9.0 to 9.9. This paper documents, for developers and researchers, the problem behind each change, the mathematics of the algorithm, and where it sits in the code, together with the tests that were run. The main items are: invariance rows of all ranks from one loop; symmetry checks and symmetrization of the structure, the Born charges and the stress; generalized Huang rows for a stressed reference; Born–Huang elastic constants, B_0' from FC3 and a rewritten quasi-harmonic model; mode Grüneisen parameters with an exact long-wave limit; standard errors of the FCs; relaxation of an unrelaxed reference, with the full Ewald energy for point charges; IR/Raman activity from computed character tables; partial refits with some ranks read; a hybrid Coulomb subtraction; elastic constants from the long-wave sound speeds; an irreducible wedge from the Laue group; OpenMP; and a constrained LASSO with exact invariances. Version 9.9 adds standard errors for LASSO fits, a noise test for the LASSO weight, the smooth SVD filter written as a per-mode quadratic penalty, an optional ridge prior, a rule that moves the SVD cut into a gap of the singular values, a refit when the measured noise exceeds the guess, and rotational invariants of FC3 and FC4. A Wigner–Seitz boundary aliasing case that motivated the gap rule is analysed, and a study of $MoTe_2$ shows how exact invariances distort the fit when the model or the data are inconsistent with them.

Contents

1 Introduction **2**

2 Invariance rows of all ranks in one loop (9.0) **3**

3 Symmetry check of Born charges and dielectric tensor (9.0) **4**

4 Range of the FC3–FC8 clusters **5**

5 Stressed reference: generalized Huang rows, and fixes from the exact-model test **5**

6 Frequencies at the exact k and the Γ limit (9.1) **6**

7 Elastic constants from the Born–Huang brackets (9.2) **7**

8 B_0' from FC3 and the quasi-harmonic model (9.2) **7**

9 Mode Grüneisen parameters (9.2) **8**

10 Standard errors of the fitted FCs, sd(rank) (9.2) **9**

11 Relaxation of an unrelaxed reference (9.3) **9**

12 IR/Raman activity and irreducible representations at Γ (9.4) **10**

13 Reading some ranks and fitting the others, `include_fc=2` (9.5) **10**

14 `born_flag=2`: hybrid Coulomb subtraction with Coulomb right-hand sides (9.5) **11**

15 Elastic constants from the long-wave sound speeds (9.5) **12**

16 Irreducible wedge from the Laue group (9.5) **12**

17 Symmetrized positions, Born charges, dielectric tensor and stress (9.6) **13**

18 OpenMP and `-O3` with unchanged results (9.7) **13**

19 Coulomb terms in the relaxation; exact Coulomb by Broyden; relax-only reruns (9.7, 9.8) **14**

20 Constrained LASSO with exact invariances (`enforce_inv=2, 3`) **14**

21 Standard errors for LASSO fits (9.9) **15**

22 The smooth SVD filter as a per-mode penalty; the ridge prior (9.9) **15**

23 The noise test for the LASSO weight (9.9) **16**

24 The SVD cut moved into a gap of the singular values (9.9) **17**

25 Refit when the measured noise exceeds the guess (9.9) **18**

26 Rotational invariants of FC3 and FC4 (9.9) **19**

27 Wigner–Seitz boundary aliasing **19**

28 Invariances with an inconsistent model: $MoTe_2$ **20**

28.1 Data consistency check 20

28.2 Huang rows with zero stress at a reference out of equilibrium 20

28.3 FC2 range 20

28.4 Rotational invariance with truncated FC3 21

28.5 Relaxation 21

29 Conclusions **21**

1 Introduction

FOCEX expands the potential energy of a crystal about a reference structure in the atomic displacements u_i^α (atom i, Cartesian direction α):

$$E = E_0 + \sum_{i\alpha} \Pi_i^\alpha u_i^\alpha + \frac{1}{2!}\sum \Phi_{ij}^{\alpha\beta} u_i^\alpha u_j^\beta + \frac{1}{3!}\sum \Psi_{ijk}^{\alpha\beta\gamma} u_i^\alpha u_j^\beta u_k^\gamma + \frac{1}{4!}\sum \chi_{ijkl}^{\alpha\beta\gamma\delta} u_i^\alpha u_j^\beta u_k^\gamma u_l^\delta + \dots \quad (1)$$

Π (FC1), Φ (FC2), Ψ (FC3), χ (FC4) and the higher ranks up to FC8 are the force constants. The force $F_i^\alpha = -\partial E/\partial u_i^\alpha$ is linear in the FCs. The space-group symmetry reduces the FCs of

each rank to a set of irreducible values; every FC is a linear combination of them. Stacking the forces of all atoms in all snapshots gives the linear system

$$\mathbf{A}\mathbf{x} \approx \mathbf{b}, \tag{2}$$

where $\mathbf{x}$ (n_x entries) are the irreducible FCs of all ranks fitted, $\mathbf{b}$ (N entries) are the DFT forces, and $\mathbf{A}$ ($N \times n_x$) holds the products of displacements.

The FCs must also satisfy linear invariance relations $\mathbf{C}\mathbf{x} = \mathbf{d}$: translational invariance (the acoustic sum rules) of every rank, rotational invariance coupling ranks n and $n-1$, and the Huang conditions on FC2. $\mathbf{d} = 0$ except where the reference is stressed or a Coulomb part was subtracted (Sections 5 and 14). The switch `enforce_inv` (`structure.params` line 5, fourth value) chooses how the two systems are combined:

- 0: the invariance rows are stacked with the force rows and the whole system is solved by SVD;
- 1: $\mathbf{x} = \mathbf{x}_p + \mathbf{K}\mathbf{y}$, with $\mathbf{K}$ an orthonormal basis of the kernel of $\mathbf{C}$ and $\mathbf{x}_p$ a particular solution; then $\min_{\mathbf{y}} \|\mathbf{A}\mathbf{K}\mathbf{y} - (\mathbf{b} - \mathbf{A}\mathbf{x}_p)\|$ by SVD (`fit_kernel_basis`, `constrained_fit.f90`). The invariances hold exactly;
- 2 and 3: a LASSO on $\mathbf{x}$ inside the same affine space, without and with a least-squares refit on the selected FCs (Section 20).

All SVD solutions use the regularized back-substitution of `svbksb` (`svd_ridge.f90`), Section 22. The relative cut is $c = \epsilon_F^2$, with ϵ_F the force error of `default.params` line 2 (default 10^{-5}, so $c = 10^{-10}$; `ios.f90`). `default.params` has 7 lines, each with a comment after its value: symmetry tolerance, force error, `icutoff` (range of the neighbour search), margin (smallest FC written), `relax_ref` (line 5), `lasso_rule` (line 6) and the ridge-prior switch `lasso_l2` (line 7). FOCEX reads only the first number of each line; lines 5–7 are optional and only the value 1 switches them on.

This paper takes the changes in order of version. Each section has a Motivation, an Algorithm and an Implementation part, and a Validation paragraph where tests were recorded. Version dates and contents are those of the manual (`docs/source/runfocex.rst`, Version history), the git tags `focex-9.0` to `focex-9.8`, and the developer notes (`CLAUDE.md` of the repository). Numbers are copied from those notes, the session notes and the README files of `FOCEX/REFERENCE/exact_model`; the 9.9 results of Sections 25 and 28 are from runs of the final 9.9 build on 6 November 2026. Version 9.9 (6 November 2026; `FOCEX/Makefile ver=9.9`, manual history entry 9.9) is to be tagged `focex-9.9`.

Many tests below use the *exact model* of `FOCEX/REFERENCE/exact_model`: an energy that is a polynomial of degree 4 in the positions (pair, three-body and four-body terms in squared distances), so FC1 to FC4 are known exactly and FC5 and higher vanish. Its reference is stressed. It is set up for fluorite CaF_2, wurtzite ZnO and monoclinic HfO_2, with 100 snapshots each.

2 Invariance rows of all ranks in one loop (9.0)

Motivation. The translational and rotational rows were built rank by rank by separate routines. The rotational routine had wrong column offsets: for rank-3 rows the FC2 part used a counter that included dropped FC2 groups (groups outside the supercell Wigner–Seitz cell, `fc2range`=0). For rank-2 rows a dropped FC2 term went to the column of the previous kept one, and the FC1 part summed the wrong group sizes. The error appears in crystals with at least two inequivalent atoms when rank 3 is fitted.

Algorithm. For rank n the first index i_1 is an atom of the primitive cell, the others are its neighbours. Fix all indices but the last (the "prefix"). Translational invariance:

$$\sum_{i_n} \Phi^{(n)}(i_1\alpha_1, \ldots, i_n\alpha_n) = 0, \qquad \text{row key } (i_1..i_{n-1}; \alpha_1..\alpha_n). \tag{3}$$

Rotational invariance, with X_i the position of atom i and ε the Levi-Civita symbol:

$$\sum_{i_n\alpha_n\mu} \Phi^{(n)}(i_1\alpha_1, .., i_n\alpha_n) X^{\mu}_{i_n} \varepsilon_{\alpha_n\mu\nu} + \sum_{k=1}^{n-1} \sum_{\gamma} \Phi^{(n-1)}(i_1\alpha_1, .., i_k\gamma, .., i_{n-1}\alpha_{n-1}) \varepsilon_{\gamma\alpha_k\nu} = 0, \tag{4}$$

with row key $(i_1..i_{n-1}; \alpha_1..\alpha_{n-1}; \nu)$. Only prefixes whose atoms $i_2..i_{n-1}$ lie within the shells of rank n are used: beyond the range of rank n the rows would force combinations of rank $n-1$ to vanish, an artefact of the truncation. Every direction combination is used. One pass over the terms of every group adds (key, column, value) triples; sorting the keys gathers each row. Repeated and proportional rows are removed afterwards (`compare2previous_lines`).

Implementation. Module `invariance_ranks` (`FOCEX/invariance_ranks.f90`), routines `set_translational_inv_constraints` and `set_rotational_inv_constraints`; the column of independent term t of group g of rank n is `column_of` (offset plus `current()`, skipping dropped FC2 terms). Added in commit `e70e986` (23 October 2026). The old rank-by-rank routines and the switches `FOCEX_INVARIANCE=old` and `FOCEX_INVARIANCE_COMPARE` were removed on 1 November 2026. The Huang rows (rank 2 only) are built elsewhere (`setup_amatrix.f90`).

Validation. Old (fixed) and new rows on the exact pair-potential model, six lattices plus trigonal: same rotational row space, residuals 10^{-16}, exact fits. Before the fix the old rotational rows had residuals 3×10^{-2} to 2×10^{-1} on the exact FCs and the constrained force error reached 0.6 eV/Å. $BaZrO_3$ (BZO) with FC3: same row space, fit 25.986 % and 25.151 % for two settings in both. The new translational rows contain the old ones and more ($MoTe_2$ 33→43, GeSe 123→220, MgO 6→7 independent rows); the fits are unchanged. DFT regression (Ge, BZO, $MoTe_2$, GeSe, MgO): new rows equal the previous results to all digits.

3 Symmetry check of Born charges and dielectric tensor (9.0)

Motivation. The BZO input had the three O Born-charge tensors cyclically permuted. With `born_flag`=2 the fit error was 25.4 %, the Γ modes were imaginary ($-131\,\mathrm{cm}^{-1}$) and the cubic degeneracy was broken. With the tensors on the right sites the fit was 3.46 %.

Algorithm. For every space-group operation $\{R|\mathbf{t}\}$ and atom k with $RX_k + \mathbf{t} = X_{k'} +$ (lattice vector), the tensors must obey

$$Z^*(k') = R\, Z^*(k)\, R^T, \qquad \epsilon = R\, \epsilon\, R^T. \tag{5}$$

The largest deviation relative to $\max|Z^*|$ (or $\max|\epsilon|$) is computed. Above 5×10^{-2} the run stops; above 10^{-4} it warns. On a stop every reassignment of the Z^* blocks among atoms of the same species is tried; if one obeys the symmetry, the reordered file is written.

Implementation. `check_born_symmetry` (`ios.f90`), called after `symmetry_init` when `born_flag`$\neq 0$. A symmetric reassignment is written to `dielectric.params.symmetric`, and the equivalent atom order of `structure.params` is printed. Since 3 November 2026 the tensors are also symmetrized below the stop threshold (Section 17).

Validation. BZO: deviation 0.50, block order 1 2 4 5 3 found. GeSe: equivalent Ge with Z_{yy} 2.598 and 2.630, deviation 1.0×10^{-2}, warning only. RbBr: 10^{-17}.

4 Range of the FC3–FC8 clusters

Motivation. The range of each rank was set only by a number of neighbour shells. Interatomic potentials are mostly pair terms plus a smaller correction, so the two-body parts of FC3 and FC4 (Ψ_{iij}, χ_{iiij}, χ_{iijj}) can reach as far as FC2. A shell count that keeps the three-body terms affordable cuts these off.

Algorithm. `structure.params` line 12 is `fc2range [fc3flag ... fc8flag]`; absent flags are 1. For a rank with flag 1 a cluster is kept if all its pairs lie within the shells of that rank (as before). For a rank with flag 0 a cluster is kept if

1. every pair of it is a kept FC2 pair (its FC2 group has a member inside the supercell Wigner–Seitz cell), and
2. all its pairs lie within the shells of the rank, or it has only two distinct atoms, or its three or more distinct atoms are collinear.

The test is applied during the enumeration, so the maps of ranks ≥ 3 hold only kept groups (complete orbits).

Implementation. `fc34_range.f90` (module `fc34_range`: `fc34_init` builds the table `pair_keep` from the FC2 groups) and `fc34_maps.f90` (`setup_maps_fc34`, `size_force_constants_fc34`, `collect_force_constants_fc34`, copies of the original routines with the rule). `focex.f90` calls `setup_maps_fc34` only if some flag is 0, so the original route is untouched. With flag 0, `clusters<rank>.dat` lists the clusters.

Validation. Without flags `maps.dat` is byte-identical to the previous code (BZO, $MoTe_2$, MgO, Ge, GeSe). The two-body FC3 distances equal the kept FC2 distances. Sizes (independent FC3, FC4) grow from 77, 25 to 534, 1722 (BZO); 95, 2 to 329, 862 (Ge); 654, 18 to 4434, 11083 (GeSe); 76, 7 to 169, 590 ($MoTe_2$); `setup_maps` for BZO from 14 s to 8 min. On the exact model with flag 0 and the FC2 range inside the Wigner–Seitz cell, Ge, BZO and GeSe fit to 0.000 % with all FCs equal to the exact ones to the 7 written digits.

Version. The files were added in commit `c9edf0a` (27 October 2026), after the `focex-9.0` tag (23 October) and before `focex-9.1`. The manual's 9.0 and 9.1 entries do not mention them.

5 Stressed reference: generalized Huang rows, and fixes from the exact-model test

Motivation. The Huang conditions assume a stress-free reference. A DFT reference is often stressed. The full test against the exact model (stressed by construction) also exposed an error in the symmetry analysis of the FC terms.

Algorithm. Define the brackets of FC2, with V the cell volume and $d = X_j - X_\tau$ the bond vector from atom τ of the primitive cell,

$$[\alpha\beta, \gamma\delta] = -\frac{1}{2V} \sum_{\tau,j} \Phi^{\alpha\beta}_{\tau j}\, d^\gamma d^\delta. \tag{6}$$

For a reference under stress σ (tensile positive) the generalized Huang conditions are

$$[\alpha\beta, \gamma\delta] - [\gamma\delta, \alpha\beta] = \delta_{\alpha\beta}\sigma_{\gamma\delta} - \delta_{\gamma\delta}\sigma_{\alpha\beta} \tag{7}$$

(derivation in `FOCEX/notes/generalized_huang.tex`). These rows have a right-hand side, so the constrained fit uses $\mathbf{x} = \mathbf{x}_p + \mathbf{K}\mathbf{y}$ with a particular solution $\mathbf{x}_p$ (`particular_solution`, `constrained_fit.f90`).

Implementation. The stress is read from the optional file `stress.params` (3×3, GPa; absent means $\sigma = 0$, the ordinary Huang rows) and echoed in the log. The Huang flag is the third value of `structure.params` line 5. The same commit `c9edf0a` (27 October 2026) also

1. fixed `unique_force_constant` (`force_constant_symmetries.f90`), which was called with the same array as input and output: when the canonical form reordered the atoms, later direction combinations of the same operation used the wrong atoms and imposed $\Phi(i\alpha, j\beta) = \Phi(i\beta, j\alpha)$ (wurtzite FC2 $yz = zy$);
2. added partial pivoting to the row reduction `xrowop2`;
3. dropped the rank-1 rotational rows (the FC1 torque), which depend on which periodic images form the cell when FC1$\neq 0$.

Validation. `huang.py`: Eq. (7) holds on the exact FC2 to 2×10^{-13}; the right-hand sides are 0.23 (ZnO) and 0.43 (HfO_2), and with the opposite sign the relation would fail by 0.45 and 0.86. Full test (`verify.py`): `enforce_inv`=0, 1, 3 reproduce the exact FCs of CaF_2, ZnO and HfO_2 (1499, 1927, 7977 unknowns) to the 7 written digits; the force fit is 0.000 % in every run. DFT regression: $MoTe_2$ (`enforce_inv`=1) 35.269 %→32.194 %, GeSe 6.111 %→5.755 % (the three fixes above, not separated).

6 Frequencies at the exact k and the Γ limit (9.1)

Motivation. Frequencies were computed at a point shifted from the requested k (to separate degenerate bands for the velocities). At $q = 0$ exactly the non-analytic (NA) term of a polar crystal is the angular average: for BZO it gave three modes at 497.8 cm^{-1} instead of 462, 462, 620.

Algorithm. The eigenvalues ω^2 are taken at k itself. Eigenvectors and Hellmann–Feynman group velocities are taken at $k + 3 \times 10^{-4}\,|G|\,\hat{u}$, with $|G|$ the cube root of the volume of the reciprocal primitive cell and $\hat{u}$ a fixed direction. A point with $|k| < 10^{-5}$ (the constant `gamma_tol`, since 9.4) is Γ: it is evaluated at $q = 10^{-6}\hat{d}$, with $\hat{d}$ the band-path segment (incoming if the previous point differs, else outgoing) or $\hat{u}$ on a mesh. The three rigid translations there get $\omega^2 = 0$ exactly and are left out of the thermodynamics.

Implementation. `freq_vel` and `gamma_limit` (`common/phonons.f90`). The old value of the translations is written to the log (`FREQ_VEL`). In the same version an FC2 group with members outside the supercell Wigner–Seitz cell no longer stops the run (`set_dynamical_matrix_std`, `fourier_fc2fit_4D`); members outside are logged once (`log_fc2_outside_ws`, `others.f90`). See Section 27 for what such groups imply.

Validation. Regression against the previous commit (Ge, BZO, $MoTe_2$, GeSe): fits identical; off Γ the frequencies change by the removed shift, at most 1.25 cm^{-1} (BZO). GeSe $\langle\gamma\rangle$ 0.7856→0.7683, the value of the old data without the Γ acoustic modes (0.7682). In 9.4 the Γ test was unified to 10^{-5}: a LaP path ended at $|k| = 2.7 \times 10^{-10}$, which `freq_vel` treated as Γ and `gamma_limit` (then 10^{-12}) did not.

7 Elastic constants from the Born–Huang brackets (9.2)

Motivation. The elastic constants were computed as $\mathrm{Sym}(A) - \mathrm{Sym}(\delta\sigma)$. On the exact relaxed ZnO model this gave C_{12} 171 instead of 221 GPa, C_{13} 137 instead of 197, C_{44} 108 instead of 78, C_{66} 146 instead of 121; the diagonal constants were right. All earlier FOCEX elastic constants, moduli, sound speeds and Debye temperatures were affected. The cause: the step from $\frac{1}{2}\sum u\Phi u$ to $-\frac{1}{4}\sum(u_i-u_j)\Phi_{ij}(u_i-u_j)$ needs $\Phi_{ij}^{\alpha\beta} = \Phi_{ij}^{\beta\alpha}$, true for central pair forces only; in general only $\Phi_{ij}^{\alpha\beta} = \Phi_{ji}^{\beta\alpha}$ holds. The part of each 3×3 block antisymmetric in its Cartesian indices is lost.

Algorithm. With the brackets of Eq. (6) and the reference stress σ,

$$c_{\alpha\beta\gamma\delta} = \mathrm{Sym}\left\{[\alpha\gamma,\beta\delta]+[\beta\gamma,\alpha\delta]-[\alpha\beta,\gamma\delta]+\tfrac{1}{2}(\delta_{\alpha\beta}\sigma_{\gamma\delta}+\delta_{\gamma\delta}\sigma_{\alpha\beta})-\delta_{\alpha\gamma}\sigma_{\beta\delta}-\delta_{\beta\delta}\sigma_{\alpha\gamma}\right\} - \frac{1}{V}\,\mathrm{Sym}(Q\Gamma Q). \tag{8}$$

Sym is the average over $\alpha\leftrightarrow\beta$, $\gamma\leftrightarrow\delta$ and $(\alpha\beta)\leftrightarrow(\gamma\delta)$. Q is the first moment of FC2 (sum over j of $\Phi_{\tau j}$ times the bond vector), the coupling of strain and internal displacements, and Γ the pseudo-inverse of the $q=0$ FC2 matrix between sublattices; the last term is the internal relaxation. The $(\alpha\beta)\leftrightarrow(\gamma\delta)$ average was added in 9.3: without it the tensor of a non-hydrostatically stressed state is not pair-symmetric (ZnO model at 15 GPa: H_{13} 51.452 and H_{31} 51.420, exact 51.436 = the average).

Implementation. `elastic_born_huang` (`dynamical.f90`), used by `mechanical`, `mechanical0` and the relaxation. Results in `mech.dat`. The stress is that of `stress.params` (point-group averaged since 9.6). Since 9.7, with Born charges on, the constants from the short-range FC2 alone go to the log only (with a warning); `mech.dat` keeps the long-wave tensor (Section 15).

Validation. `REFERENCE/exact_model/bprime/elastic_check.py` against the model's strain Hessian: within 6×10^{-10} GPa for relaxed ZnO, CaF_2 and CaF_2 under 14.5 GPa; 0.2 GPa on the stressed ZnO and HfO_2 references, whose residual forces are not in the formula. Ge (DFT): C_{11}, C_{12}, C_{44} = 121.1, 55.0, 53.9 GPa; the old code gave C_{12} 68.0 > C_{44} 47.4.

8 B_0' from FC3 and the quasi-harmonic model (9.2)

Motivation. The old QHA took $V(T)$ as a Boltzmann-weighted average over strains with an artificial temperature, the expansion coefficient as a secant, $B(T) = B_0(1-\epsilon(4+3\langle\gamma\rangle))$, and a static energy quadratic in ΔV (equivalent to $B_0' = -1$).

Algorithm. B_0'. Let η_1, u_1 be the strain and internal relaxation per unit pressure (path p_1). Deform the crystal by $\pm\lambda p_1$: FC2 becomes $\Phi \pm \lambda\,\delta\Phi$ with $\delta\Phi_{\tau j} = \sum_k \Psi_{\tau jk}(S_k - S_\tau)$, $S_j = \eta_1 X_j + u_1(\tau_j)$, the bond vectors and the pressure change accordingly, and the relaxed bulk modulus $B(\pm\lambda)$ is computed from the elastic constants of the deformed crystal. Then

$$B_0' = \frac{B(+\lambda) - B(-\lambda)}{2\lambda}, \qquad \lambda = 10^{-4}\ \mathrm{eV/\AA^3}. \tag{9}$$

FC4 does not enter. Without FC3, $B_0' = 1 + 2\gamma$ with Leont'ev's $\gamma = 1.5(3-4x^2)/(1+2x^2)$, $x = v_t/v_l$ (Dugdale–MacDonald). *QHA.* $F(V,T) = E_{\mathrm{st}}(V) + F_{\mathrm{ph}}(V,T)$ on 61 isotropic strains from −5 % to +10 %, with $\omega(V)$ from the mode Grüneisen parameters. The static part is Murnaghan's,

$$P_{\mathrm{st}}(V) = P_0 + \frac{B_0}{B_0'}\left[\left(\frac{V_0}{V}\right)^{B_0'} - 1\right], \qquad E_{\mathrm{st}} = -\int P_{\mathrm{st}}\,dV, \tag{10}$$

with B_0 the Reuss modulus and P_0 the residual pressure. $P = -\partial F/\partial V$ is summed analytically; $V(T)$ is the root of $P = 0$ (local cubic fit); $B_T = -V\,dP/dV$ there; $\alpha_V = (\partial P/\partial T)_V/B_T$ =

$d\ln V/dT$; the linear coefficient is $\alpha_V/3 = d\ln L/dT$; $C_p = C_v + VTB_T\alpha_V^2$. Harmonic-only fits use Leont'ev's γ for all modes plus the term $P = NkT/V$ ($F = -NkT\ln(V/V_0)$).

Implementation. `pressure_derivative_b` and `QHA_free_energy_vs_strain` (`dynamical.f90`); `thermo_QHA.dat` (column 11 $B_T(T)/B_T(T{<}8\,\mathrm{K})$ since 9.3, columns 12, 13 B_T and B_0). The wrong B_0' of `cubic_elastic` is no longer computed (9.7). With Born charges the QHA takes B_0 and the sound speeds from the long-wave $E = 0$ tensor (9.7). The FC2-only rule $1 + 2\gamma$ is commit `3b638a7`, contained in the tag `focex-9.2` (`acf1968`).

Validation. FOCEX on relaxed, cubic-truncated exact-model inputs (`bprime/mkrelaxed.py`): ZnO $B = 288.001543\,\mathrm{GPa}$, $B' = -0.157427$; CaF_2 587.376258 and 0; equal to full relaxations under pressure. Evaluating the second-order term with the plain bracket sum instead gives $B' = -0.064$ for ZnO (wrong, same cause as Section 7). α equals the numerical $d\ln L/dT$ of the table to 0.8 % (ZnO model with FC3) and 0.7 % (FC2 only).

9 Mode Grüneisen parameters (9.2)

Motivation. FOCEX built the matrix W below with the phase $e^{iq\cdot(R+\tau'-\tau)}$, while its eigenvectors come from a dynamical matrix with $e^{iq\cdot R}$; γ was right only at $q = 0$ and off by up to 3.2 on the exact HfO_2 model. THERMACOND's routine had no internal relaxation. At Γ the acoustic γ is 0/0; evaluated at small q it loses digits as $\varepsilon_{\mathrm{mach}}/q^2$.

Algorithm. Under an isotropic strain ϵ with relaxed internal coordinates the atoms move by $\epsilon\, y_{R\tau}$, $y_{R\tau} = R + \tau + \xi_\tau$, $\xi_\tau = -\sum_{\tau'} \Gamma_{\tau\tau'}\,\mathrm{Tr}\,Q_{\tau'}$. Then

$$\gamma_s(q) = -\frac{\langle e_s|W(q)|e_s\rangle}{6\,\omega_s^2(q)}, \qquad W_{\tau\alpha,\tau'\beta}(q) = \sum_{R'}\sum_{k} \frac{\Psi^{\alpha\beta\gamma}_{0\tau,R'\tau',k}\, y_k^\gamma}{\sqrt{m_\tau m_{\tau'}}} e^{iq\cdot R'}, \tag{11}$$

with the same phase as the dynamical matrix D. Along $\hat{q}$ write $D = D_0 + qD_1 + q^2D_2$ and $W = W_0 + qW_1 + q^2W_2$. With T the mass-weighted rigid translations, $P = 1 - TT^\dagger$ and $G = (PD_0P)^+$,

$$A = T^\dagger D_2 T - T^\dagger D_1 G D_1 T, \tag{12}$$

$$B = T^\dagger W_2 T - T^\dagger W_1 G D_1 T - T^\dagger D_1 G W_1 T + T^\dagger D_1 G W_0 G D_1 T, \tag{13}$$

$$\gamma_s(\hat{q}) = -\frac{b_s}{6a_s}, \tag{14}$$

with a_s, u_s the eigenvalues and eigenvectors of A and $b_s = u_s^\dagger B u_s$ (within a degenerate subspace, the eigenvalues of B there). W_0, W_1, W_2 are exact sums over the FC3 terms; D_0, D_1, D_2 come from the code's own D (NA term included) at $\pm h, \pm 2h, \pm 3h$, $h = 10^{-3}|G|$. The limit is used for the three translations at Γ and for the acoustic modes at $0 < |k|V^{1/3} < 0.05$. The eigenvectors at k are those of $D(k)$, with W diagonalized inside each degenerate set (tolerance $10^{-7}\max\omega^2$).

Implementation. Module `gruneisen_rules` (`common/gruneisen.f90`): `mode_gruneisen`, `gruneisen_longwave`, `internal_strain_trace`; shared with THERMACOND, each code passing its FC3 terms (`w_terms_set`) and its dynamical matrix. Log section `GRUNEISEN_LONGWAVE`. Derivation: `generalized_huang.tex`, Part III.

Validation. Exact HfO_2 model: the direct values approach the limit as q^2 down to $q \simeq 3\times 10^{-4}\,\text{Å}^{-1}$; at $q = 10^{-3}$ for CaF_2, ZnO, HfO_2, four directions, with and without relaxation, $\max|\gamma^{\mathrm{direct}} - \gamma(\hat{q})| \le 8\times 10^{-6}$. FOCEX on the cubic-truncated HfO_2 model, ranks 1–3: equal to the exact γ to the printed 5×10^{-5} at all 519 path points off Γ; THERMACOND to its 4 digits; Γ limit -1.691288, -1.344395, -0.658566 in both.

10 Standard errors of the fitted FCs, sd(rank) (9.2)

Motivation. For `enforce_inv`=1 the errors were read from an unset array; for 0 they were scaled by $\sqrt{c}$ and divided again by $\sqrt{N}$.

Algorithm. With the SVD $\mathbf{AK} = \mathbf{UWV}^T$ (singular values w_k, cut $w_c = c\,w_{\max}$), the residual variance per row $s^2 = \|\mathbf{AKy} - \mathbf{b}\|^2/(N - n_K)$ and the filter $f(w) = w^4/(w^6 + w_c^6)$ ($= 1/w^2$ well above the cut),

$$\sigma_j = s\,\sqrt{\left[\mathbf{K}\mathbf{V} f(\mathbf{W})\mathbf{V}^T\mathbf{K}^T\right]_{jj}}. \tag{15}$$

For `enforce_inv`=0 the same holds with $\mathbf{K} = 1$ and the stacked matrix. sd(rank) is the mean σ of the irreducible FCs of that rank.

Implementation. `svd_set` (`svd_ridge.f90`) returns $\mathbf{V}\,\mathrm{diag}(s\sqrt{f(w)})$, which `fit_kernel_basis` maps by $\mathbf{K}$. The log lists every irreducible FC with its σ (`write_independent_fcs`). In 9.9 the array `sd` was enlarged from 4 to 8 entries (the writer fills 8).

Validation. Exact ZnO model, ranks 1–3, Gaussian force noise 10^{-3}: estimated $s = 0.9948 \times 10^{-3}$, 1.0005×10^{-3}, 0.9776×10^{-3} (two seeds; `enforce_inv`=1, 0); rms of $(x - x_{\mathrm{exact}})/\sigma$ 1.03 for FC2, 0.84–0.90 for FC3; 72–76 % within one σ. A jackknife (5 refits leaving out 20 of the 100 snapshots): ratio to the SVD σ, median 1.04 (FC2), 0.91 (FC3), rms ratio 0.96. Open: why the FC3 errors are 10–15 % below the estimate.

11 Relaxation of an unrelaxed reference (9.3)

Motivation. A DFT reference may carry residual forces (FC1) and stress. Downstream quantities (phonons, elastic constants, QHA) should be those of the relaxed structure.

Algorithm. The static displacement is $S_{R\tau} = \eta\,(R + \tau) + u_{0,\tau}$, with η the strain and u_0 the internal shifts (u_0 of atom 1 held at 0). The Taylor energy Eq. (1) with $u = S$ is minimized over (η, u_0). The FCs are re-expanded about the minimum,

$$\Pi' = \Pi + \Phi S + \tfrac{1}{2}\Psi SS + \tfrac{1}{6}\chi SSS, \qquad K = \Phi + \Psi S + \tfrac{1}{2}\chi SS, \qquad \Psi' = \Psi + \chi S, \tag{16}$$

with $S_j - S_\tau$ in place of S_j (equal by the sum rules and independent of the image chosen). Stage 1 relaxes u_0 in the reference cell. Stage 2 takes Newton steps in the six symmetric strain components, with u_0 relaxed at each point of the path (a force-free path). Along the path $F(t) = 1 + t\Delta$ the second Piola–Kirchhoff stress S is integrated by

$$\frac{dS_{ab}}{dt} = J\,F^{-1}_{aj}F^{-1}_{bk}F^{-1}_{cl}F^{-1}_{dm}\,c^{\mathrm{rel}}_{jklm}\,\dot{E}_{cd}, \qquad \dot{E} = \tfrac{1}{2}(\Delta^T F + F^T\Delta), \quad J = \det F, \tag{17}$$

with c^{rel} the relaxed elastic tensor of the current state (Section 7). The rotation is removed from the accumulated deformation gradient after each step by polar decomposition. The ALATDYN paper's strain Hessian $A = -\frac{1}{2}\sum K\,d\,d$ is not used: it is the symmetrized form of Section 7 and is wrong off the Cauchy case.

Implementation. `relax_reference` (`FOCEX/relax_reference.f90`), switched on by `default.params` line 5 = 1 (any other value is off). After it, the lattice, positions and grids are strained by $1 + \eta$ and shifted by u_0; the re-expanded FCs are projected on the irreducible FCs and used downstream; `stress.params` is then read as 0. The unrelaxed fit is kept as `fcN_unrelaxed.dat`, `fcN_irr_unrelaxed.dat`, `lat_fc_unrelaxed.dat`; `POSCAR_relaxed` is written. Since 9.7 the average force per atom is removed from FC1 before relaxing ($\sum_\tau \Pi_\tau = 0$), and the fractional translations of the space group are re-derived for the relaxed cell.

Validation. Exact ZnO model (stressed 14.9 GPa, FC1 0.032 eV/Å, ranks 1–4, `enforce_inv`=1) against the model's own minimum: strain 7.8×10^{-10}, u_0 1.3×10^{-9} Å, FC2 2.7×10^{-6} eV/Å^2, FC3 5×10^{-7}; residual stress 2×10^{-10} GPa; 3 strain steps. With ranks 1–3 the strain is off by 10^{-5} (FC4 missing). With the paper's A the first version gave ZnO strain -0.01831 against the exact -0.017837. Monoclinic HfO_2 model (`HfO2r`, 12 atoms, no charges, 4 November): strain 1.3×10^{-9}, u_0 2.9×10^{-9} Å, FC2 7.4×10^{-8} (of 6.55), FC3 5.1×10^{-8} (of 2.37); stress after 8.7×10^{-15} GPa.

12 IR/Raman activity and irreducible representations at Γ (9.4)

Motivation. Users need the symmetry labels and the IR/Raman activity of the Γ modes. The character tables are computed rather than looked up, so that the routine also serves other problems (electrons, little groups).

Algorithm. At $q = 0$ the modes transform under the space group modulo lattice translations. Operation $g = \{S|\mathbf{t}\}$ takes atom k to $\mathrm{img}(k, g)$ and acts on a displacement pattern by $\Gamma(g)_{3(\mathrm{img}-1)+a,\,3(k-1)+b} = S_{ab}$. For a degenerate set with eigenvectors E (of D at $q = 0$, NA term direction-averaged) the characters are $\chi(g) = \mathrm{Tr}(E^\dagger \Gamma(g) E)$, and

$$n_{\mathrm{IR}} = \frac{1}{|G|} \sum_g \chi^*(g)\, \mathrm{Tr}\, S_g, \qquad n_{\mathrm{Raman}} = \frac{1}{|G|} \sum_g \chi^*(g)\, \frac{(\mathrm{Tr}\, S_g)^2 + \mathrm{Tr}\, S_g^2}{2}, \tag{18}$$

the multiplicities of the vector and of the symmetric second-rank tensor representation; parity $\chi(i)/\dim$. Character tables by the class-multiplication method: with classes K_i of sizes h_i and $K_i K_j = \sum_k c_{ijk} K_k$, the vectors $w_k = h_k \chi(K_k)/d$ of an irrep of dimension d satisfy $M_i w = w_i w$, $(M_i)_{jk} = c_{ijk}$. The matrices $D^{-1/2} M_i D^{1/2}$ ($D = \mathrm{diag}(h)$) are normal; a random real combination of their Hermitian and anti-Hermitian parts is diagonalized; $\chi_k = d w_k / h_k$ with $d^2 = |G| / \sum_k |w_k|^2 / h_k$.

Implementation. `gamma_mode_symmetry` (`dynamical.f90`) writes `gamma_modes.dat`; module `point_group_rules` (`common/point_group.f90`): `pg_setup`, `pg_reduce`, `pg_write_table`, `pg_chi_vector`, `pg_chi_sym2`. Labels: A/B, E, T, g/u, $'/''$, a serial number only when needed; classes in a fixed order so that the labels do not depend on the order of the operations. One symmetry tolerance `tol_sym`= 10^{-3} Å (`common/params.f90`) for accepting operations and finding atom images. Acoustic modes by their translation weight.

Validation. `REFERENCE/point_groups/test_point_groups.f90`: all 32 point groups (order, classes, dimensions, orthogonality); O_h T_{1u} = 3 1 0 −3 −1 −1 −1 0 1 1 as the standard table. `gamma_modes.dat`: ZnO $2A_1 + 2B_2 + 2E_2 + 2E_1$; HfO_2 $9A_g + 9A_u + 9B_g + 9B_u$ (acoustic $A_u + 2B_u$); LaP $2T_{1u}$. LaP IR strength 0.962 = 3×0.3207 of the dielectric table.

13 Reading some ranks and fitting the others, `include_fc`=2 (9.5)

Motivation. One may want to keep FC2 from a dense harmonic data set and fit only FC3, or rerun a part of the analysis without refitting everything.

Algorithm. Partition $\mathbf{x} = (\mathbf{x}_{\mathrm{fit}}, \mathbf{x}_{\mathrm{read}})$. Force rows and invariance rows become

$$\mathbf{A}_{\mathrm{fit}} \mathbf{x}_{\mathrm{fit}} = \mathbf{b} - \mathbf{A}_{\mathrm{read}} \mathbf{x}_{\mathrm{read}}, \qquad \mathbf{C}_{\mathrm{fit}} \mathbf{x}_{\mathrm{fit}} = \mathbf{d} - \mathbf{C}_{\mathrm{read}} \mathbf{x}_{\mathrm{read}}. \tag{19}$$

The Huang rows contain FC2 only and are dropped when FC2 is read. The irreducible values of a read group g are the least-squares solution of $\sum_t M_{t,t'} x_{t'} = f_t$ over its terms t (absent terms $f_t = 0$); the residual is logged.

Implementation. `include_fc` is `structure.params` line 4. `read_fcs` (`ios.f90`) reads `fcN.dat` and the earlier `lat_fc.dat`, and matches atoms by (sublattice, cell), whatever the numbering. Module `given_fcs` (`xfull`, `full_col`); force-row right-hand side in `setup_amatrix.f90`; invariance right-hand side and `merge_given_fcs` in `invariance_ranks.f90`. After the fit the read ranks count as fitted, so all output is that of a full fit. `fcN.dat` and `fcN_irr.dat` carry 8 significant digits (format `g15.8`) since 3 November 2026. In 9.8: if every included rank is read, the force data, invariance rows and fit are skipped (relax-only rerun, Section 19).

Validation. Cubic-truncated HfO_2 model: full fits byte-identical; reading FC3, or FC1+FC2, gives the rest to the printed digits; reading FC2 gives FC3 to 10^{-4}. Reading FC2 at 7, 8 and 15 digits gives FC3 relative errors 1.1×10^{-4}, 5.3×10^{-5}, 2.4×10^{-9}: the error is the rounding of the input. A random relabelling of atoms 13..6500 in `lat_fc.dat` and `fcN.dat` gives byte-identical outputs.

14 `born_flag`=2: hybrid Coulomb subtraction with Coulomb right-hand sides (9.5)

Motivation. With `born_flag`=2 FOCEX subtracts a long-range Coulomb force from the DFT forces, fits the short-range remainder, and adds the dipole term back to the dynamical matrix. With anisotropic Z^* the harmonic part of the subtracted force was not the term added back (per Zn–O pair xz 0.170 and zx 0.188 against 0.179). The invariance rows on the short-range part used the total residual forces and stress, while the dipole FCs alone carry their own (ZnO: deviatoric stress $(-0.108, -0.108, +0.216)$ GPa).

Algorithm. The subtracted force is

$$F_{\mathrm{sub}} = F_m(x) - F_m(x_0) - (\Phi_C - \Phi_m)\, u, \tag{20}$$

with F_m the Ewald force of the pair model with couplings $c_{ij} = \mathrm{Tr}(Z_i Z_j^T)/3$ in the dielectric ϵ; Φ_C the Coulomb dipole FCs ($e^2 Z_t T(R + x_u - x_t) Z_u^T$, $T = -\nabla\nabla\phi_\epsilon$, $E = 0$), equal to the term added back to D; and Φ_m the harmonic part of the pair model. The harmonic part subtracted is then exactly Φ_C. The fit sees $\mathbf{x}_{\mathrm{fit}} = \mathbf{x} - \mathbf{x}_{\mathrm{sub}}$, so every included invariance row gets the right-hand side $-\mathbf{C}\mathbf{x}_{\mathrm{sub}}$: Huang rows $-(S_C(\alpha\beta, \gamma\delta) - S_C(\gamma\delta, \alpha\beta))$; rank-2 rotational rows $-\sum_{a\mu} M(i_1; \alpha_1, a, \mu)\varepsilon_{a\mu\nu}$ with $M = \sum_j \Phi_C(i_1\alpha_1, ja)(X_j - X_{i_1})_\mu$; rank-3 rotational rows from the FC3 of the pair model. The lattice sums are conditionally convergent; they come from the small-q expansion of the $E = 0$ Ewald sum (central differences with Richardson extrapolation).

Implementation. `hybrid_harmonic`, `subtract_coulomb_force` (`ewald_born.f90`); module `coulomb_lr` (`coulomb_lr.f90`: `lr_brackets`, `lr_moments`, `lr_rot3_pair`, `lr_rot3_onsite`); Huang right-hand side in `setup_amatrix.f90`, rotational in `invariance_ranks.f90`. The `mech.dat` Huang and rotational checks include the same terms. No switch: the right-hand sides are always applied when the rows are included. In 9.6 the correction $\Phi_C - \Phi_m$ is computed for each supercell (it was computed for the first only; GaN with two supercells gave fits of 2.98, 3.24, 712, 4.95 % on identical runs; after the fix 1.719 %). `born_flag`> 2 prints a warning (9.2): use 0 or 2.

Validation. Exact SR model plus point charges in $\epsilon = 5$ (`coulomb/mkcoulomb.py`): ZnO ranks 1–4 exact to the printed digits, fit 0.000 %. Cubic HfO_2 ranks 1–3: fit 0.000 % with the right-hand sides; 0.095 % without the Huang part, 0.510 % without the rotational part, 0.523 % without both. User's ZnO (DFT, `born_flag`=2): fit 1.720 %→1.269 % (hybrid)→1.232 % (rank-3 rotational right-hand side).

15 Elastic constants from the long-wave sound speeds (9.5)

Motivation. The bracket route of Section 7 uses the fitted (short-range) FC2 only; with `born_flag`=2 it misses the Coulomb part (user's ZnO: $C_{33} = -58$, $C_{44} = -33$ GPa). The long-wave limit of the full dynamical matrix contains it.

Algorithm. Along 13 directions $\hat{q}$ (axes, face and body diagonals) the acoustic matrix $A(\hat{q})$ ($\omega^2 = q^2 \operatorname{eig} A$, optical modes folded out exactly as in Section 9) gives, with ρ the mass density,

$$\rho A_{ik} = \sum_{jl} B_{(ik),(jl)} \hat{q}_j \hat{q}_l = (C_{ijkl} + \delta_{ik} s_{jl}) \hat{q}_j \hat{q}_l. \tag{21}$$

The 36 brackets B are fitted by least squares (78 equations, rank 36). The stressed Huang relation $B_{ab,cd} - B_{cd,ab} = \delta_{ab} s_{cd} - \delta_{cd} s_{ab}$ gives the deviatoric stress (15 equations, 5 unknowns; the pressure is invisible to long waves); C follows from $B - \delta_{ik} s_{jl}$, with the pressure taken from FOCEX's residual stress. With Born charges two tensors are given: $E = 0$ (the macroscopic-field part of the $G = 0$ term left out, switch `ewald_no_macro` in `common/ewald.f90`) and with the NA term.

Implementation. `elastic_longwave` (`dynamical.f90`), `acoustic_tensor` (`common/gruneisen.f90`); `mech.dat` section `ELASTIC_LONGWAVE`; isotropic moduli of every tensor by `moduli_from_c` (table `ELASTIC_MODULI`). For one atom per cell the optical terms are skipped (they divided by zero).

Validation. `REFERENCE/exact_model/longwave`: deviatoric stress to 10^{-13} GPa and C to 5×10^{-10} GPa (ZnO, HfO_2 stressed, CaF_2); FOCEX on the relaxed ZnO model equals the strain Hessian to 10^{-3} GPa. User's ZnO without charges: equal to the bracket route within 0.06 GPa. Argon: 3.074, 1.540, 1.537 GPa by both routes.

16 Irreducible wedge from the Laue group (9.5)

Motivation. FOCEX reduced k-meshes with the point group of the lattice. For a crystal of lower symmetry than its lattice this makes inequivalent points equivalent: CdI_2 on an 8×8×8 mesh had 50 instead of 65 irreducible points.

Algorithm. The operations are the rotations of the space-group operations (screw axes and glide planes included) and their products with inversion (time reversal, $\omega(-k) = \omega(k)$). Each unassigned mesh point starts a star; every member is mapped onto its mesh point modulo G; the weight is the number of mesh points in the star. The representative is the member, folded into the Wigner–Seitz cell, with the largest $\hat{u} \cdot k$ for the fixed generic direction $\hat{u} = (0.9, \sqrt{2} - 1, \sqrt{5} - 2)$: $\{k : \hat{u} \cdot k \geq \hat{u} \cdot Rk \ \forall R\}$ is a fundamental domain, so the irreducible points fill one wedge.

Implementation. Module `ibz_rules` (`common/ibz.f90`): `laue_operations`, `ibz_reduce`, shared with THERMACOND (whose algorithm it is); the `get_weights` routines of both codes are wrappers. The ballistic conductance per Å^2 (`conductance.dat`) uses mesh values reconstructed from the wedge through the Laue operations.

Validation. Against spglib `get_ir_reciprocal_mesh` (time reversal) for 9 structures × 4 meshes: same stars, weights and wedge. FOCEX regression (Ge, BZO, $MoTe_2$, GeSe): thermodynamic outputs within 1 in the last printed digit; THERMACOND Ge byte-identical; CdI_2 end to end in both codes equal to spglib.

17 Symmetrized positions, Born charges, dielectric tensor and stress (9.6)

Motivation. DFT positions of GaN and ZnO were off the space group by 3–8 $\times 10^{-5}$ Å, below `tol_sym`. The operations were found and the FCs symmetrized exactly, but the Huang and rotational rows use the actual positions: rows that vanish by symmetry got coefficients 10^{-3} of the largest and acted as spurious constraints. GaN: $C_{66} = -7288$ GPa; ZnO (`enforce_inv`=1): fit 58.7 %.

Algorithm. Repeat until the positions change by less than 10^{-13} Å (at most 50 passes):

$$\mathbf{t}_g \leftarrow \mathbf{t}_g - \big\langle R_g X_k + \mathbf{t}_g - L_{gk} - X_{\mathrm{img}(k,g)} \big\rangle_k, \qquad X_j \leftarrow \frac{1}{|G|} \sum_g \big(R_g X_k + \mathbf{t}_g - L_{gk} \big),\ \mathrm{img}(k,g) = j, \quad (22)$$

with L_{gk} the lattice vector that brings the image back; the centroid is kept. Tensors: $Z(k) \leftarrow |G|^{-1} \sum_g R_g^T Z(\mathrm{img}(k,g)) R_g$, $\epsilon \leftarrow |G|^{-1} \sum_g R_g \epsilon R_g^T$, and the same for the stress.

Implementation. `symmetrize_positions` and `pointgroup_average2` (`force_constant_symmetries.f90`), called from `focex.f90`; `symmetry_init` is rerun. The symmetric positions enter the rotational rows, the Huang rows and the relaxation; displacements stay FORCEDISP minus POSCAR, forces as read.

Validation. GaN: C_{66} −7288 → 110 GPa, no imaginary modes. ZnO: 58.7 %→1.30 %; `enforce_inv` 0, 1, 2 agree within 4 GPa. Exact monoclinic HfO_2 with positions perturbed by 5×10^{-5} Å: 9.5 fits 92.5 % and 1429 %, 9.6 0.000 % (FC2 1.7×10^{-6} relative). Exact ZnO plus Coulomb: tensor change 4×10^{-16}, FCs equal to 10^{-9} relative.

18 OpenMP and `-O3` with unchanged results (9.7)

Motivation. The relaxation of the monoclinic HfO_2 model took 7843 s, dominated by the SVDs.

Algorithm. Rule: an OpenMP loop must give the numbers of the serial code. Every parallel loop is over independent outputs, each computed by one iteration with its sums in the old order. In `svdcmp` the right Householder step is done by column blocks with OpenMP over columns. One SVD gives both the kernel and the particular solution (`kernel_and_particular`; there were two SVDs of the same matrix). Batching the QR rotations was tried and was slower.

Implementation. `svd_ridge.f90`; threads from `common/omp_threads.f90`: `FOCEX_NUM_THREADS` if set; 1 under an MPI launcher; `SLURM_CPUS_PER_TASK`; else cores minus max(1, cores/5). Printed in the log and `times.dat`. Default build `-O3 -fopenmp` (`make DEBUG=1`: `-O2 -fcheck=all -fbacktrace`). The U and V dumps are written only with verbose (2.1 GB and 572 MB for HfO_2).

Validation. `svdcmp`: bit-identical on a scratch driver (3000×1200, 6000×2400) and on all HfO_2 outputs (`cmp`). `-O3` against `-O2 -fcheck=all`: FCs, frequencies, thermodynamics identical (4 regression cases, HfO_2). HfO_2 relaxation 7843 s → 6520 s (`-O3`) → 1596 s (8 threads). Left serial: rotational rows (735 s), translational rows (247 s), force rows.

19 Coulomb terms in the relaxation; exact Coulomb by Broyden; relax-only reruns (9.7, 9.8)

Motivation. With `born_flag`=2 the fitted FCs are short-range only; the relaxation of 9.3 had no Coulomb term. In 9.7 the Coulomb FCs entered to second order (lattice sums of Φ_C at $q = 0$, $E = 0$). A second-order Coulomb energy is not rotationally invariant; on the exact HfO_2 model with point charges the strain was off by 1.4×10^{-4} and u_0 by 6.4×10^{-3} Å.

Algorithm (9.8). For $Z^* = z\mathbb{1}$,

$$E(X) = E_{\mathrm{SR}}(X) + E_C(X) - E_C(0), \qquad X = (\eta_1..\eta_6,\ u_0 \text{ of atoms } 2..), \tag{23}$$

with E_C the Ewald energy of the pair model ($c_{ij} = \mathrm{Tr}(Z_i Z_j^T)/3$, ϵ fixed) at the deformed cell and positions, and E_{SR} the Taylor series of the fitted FCs with FC1 $-F_C(0)$ and $\sigma_0 - \sigma_C(0)$. That split is required: with the total FC1 and stress the short-range part is not rotationally invariant. $G(X) = \partial E/\partial X = 0$ is solved by modified Broyden mixing on $f = -H_0^{-1}G$, H_0 the Hessian at the reference with Φ_C. $\partial E_{\mathrm{SR}}/\partial u_0 = \Pi'$. $\partial E_{\mathrm{SR}}/\partial \eta$: the stress formula holds only at force-free states, so the short-range stress is integrated along the short-range force-free path $u_0^*(\eta)$ (Section 11), giving $V\sigma F^{-T}$, and then carried from u_0^* to u_0 at fixed η by $\int Q^T du$ (Simpson, exact since Q is quadratic). For anisotropic Z^* the second-order route is kept, with a warning.

Implementation. `relax_broyden`, `coulx_setup`, `gradfun`, `sr_path_to`, `coul_energy` in `relax_reference.f90`; `ewald_cell_efw` (`ewald_cell.f90`: energy, forces, analytic $dE/d\eta$); `bro90` (`broyden.f90`, copied from SCOP8). The log checks the Ewald derivative against 4-point differences (9×10^{-10}) and its independence of the Ewald parameter (2×10^{-13}). Relax-only rerun: set `include_fc`=2 for every rank and copy `fcN_unrelaxed.dat`→`fcN.dat`, `lat_fc_unrelaxed.dat`→`lat_fc.dat`.

Validation. Exact HfO_2 model with point charges (`compare_relaxed.py`): strain 2.7×10^{-9}, u_0 6.5×10^{-9} Å, FC2 1.1×10^{-7}, FC3 9.4×10^{-8} (second-order route: 1.4×10^{-4}, 6.4×10^{-3}, 5.8×10^{-3}, 1.6×10^{-2}); 14 Broyden steps. Relax-only rerun 219 s instead of 33 min, strain equal to the full run to 10^{-7}. Regression Ge, BZO, $MoTe_2$, GeSe byte-identical to 9.7.

20 Constrained LASSO with exact invariances (`enforce_inv`=2, 3)

Motivation. The former LASSO penalized the kernel coefficients $\mathbf{y}$; the columns of $\mathbf{K}$ are dense mixtures, so sparse FCs are dense in $\mathbf{y}$. On an exact pair model it gave 98/80/376 non-zero FC2/FC3/FC4 for 11/20/43 true ones, and FC4 wrong.

Algorithm. With w_j the rms of column j of $\mathbf{A}$ (floored at $10^{-8} \max w$),

$$\min_{\mathbf{x}} \ \frac{1}{2N}\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \lambda \sum_j w_j |x_j| \quad \text{subject to} \quad \mathbf{x} = \mathbf{x}_0 + \mathbf{K}\mathbf{y}. \tag{24}$$

Let $\mathbf{z} = \mathbf{W}\mathbf{x}$ and $\mathbf{Q}$ an orthonormal basis of the range of $\mathbf{W}\mathbf{K}$, $\mathbf{z} = \mathbf{z}_0 + \mathbf{Q}\mathbf{t}$. Then

$$\min_{\mathbf{t}} \ \tfrac{1}{2}\mathbf{t}^T\mathbf{G}\mathbf{t} - \mathbf{c}^T\mathbf{t} + \lambda\|\mathbf{z}_0 + \mathbf{Q}\mathbf{t}\|_1, \qquad \mathbf{G} = \tilde{\mathbf{A}}^T\tilde{\mathbf{A}},\ \mathbf{c} = \tilde{\mathbf{A}}^T\mathbf{b}/\sqrt{N},\ \tilde{\mathbf{A}} = \mathbf{A}\mathbf{W}^{-1}\mathbf{Q}/\sqrt{N}. \tag{25}$$

For $\lambda \geq \lambda_{\max} = \|\mathbf{Q}\mathbf{c}\|_\infty$ (and $\mathbf{z}_0 = 0$) the solution is $\mathbf{t} = 0$. The problem is solved as a quadratic program by a primal–dual interior-point method (Mehrotra predictor–corrector): the Newton step reduces to $(\mathbf{G} + \mathbf{Q}^T\mathbf{D}\mathbf{Q})\delta\mathbf{t} = \text{rhs}$ with $D_i = 4A_iB_i/(A_i + B_i)$, positive definite, solved by

Cholesky. The support is $S = \{j : |z_j| > 10^{-8} \max |z|\}$. The refit (`enforce_inv`=3) is least squares over $\mathbf{t}$ in the null space of $\mathbf{Q}_{S^c,:}$ (Householder QR with column pivoting). λ is chosen by 5-fold cross-validation on interleaved rows over 40 log-spaced values from $\lambda_{\max}$ to $10^{-4}\lambda_{\max}$; while the minimum is at the end the grid is extended by two decades, down to $10^{-8}\lambda_{\max}$; ties within $10^{-13}\langle b^2 \rangle$ take the largest λ.

Implementation. Module `lasso_constrained` (`lasso_constrained.f90`): `classo_setup`, `ipm_classo`, `refit_support`, `classo_path`, `classo_cv_res`, `classo_cv`, `resolved_kernel`, `null_space_qr`; called from `lasso_kernel_basis` (`constrained_fit.f90`). `FOCEX_LASSO=old` selects the old method. Table in `lasso.dat`. Added in commit `c9edf0a` (27 October 2026). Background and earlier tests: `FOCEX/notes/constrained_lasso.tex` (24 October; its hard cut and tie tolerance 10^{-10} were replaced later).

Validation. Exact model (`REFERENCE/exact_model/README.md`): `enforce_inv`=3 selected exactly the true support (CaF_2 18/61/190, ZnO 2/47/229/553) and the FCs to the written digits; `enforce_inv`=2 lost no true FC, with FC4 shrinkage $\leq 3.6 \times 10^{-4}$ relative. The old grid floor $10^{-4}\lambda_{\max}$ had lost 76 true FCs on ZnO. Cubic HfO_2, ranks 1–3, 1566 unknowns: 24 min, no true FC lost, FC1–3 errors 0, 7×10^{-13}, 4.5×10^{-11} relative. Interior point against cvxpy/Clarabel: objective to 10^{-11}.

LASSO against `enforce_inv`=1 on DFT data (MgO, 9.9 build). Same data and setup as the MgO run of Section 25. `enforce_inv`=3 kept 74 of 556 FCs, fit 4.26 % (default line 2: 75 kept, 5.69 %), interior point `info`=2 (not converged at some λ); `enforce_inv`=1: fit 0.17 %. The phonons are the same (T_{1u} 534.9, maximum 748.9 cm^{-1}, none unstable); Grüneisen median 1.59 (`enforce_inv`=3) against 1.73 (1). Here LASSO removed FCs the data support; the improvement over the earlier MgO runs came from the SVD cut (Sections 24, 25), not from LASSO.

21 Standard errors for LASSO fits (9.9)

Motivation. With `enforce_inv`=2, 3 the log printed sd= 0 (not estimated).

Algorithm. Take the support S as in Section 20. The FCs with the invariances exact and $x_j = 0$ off S are $\mathbf{x} + \mathbf{KNz}$, $\mathbf{N}$ an orthonormal basis of the null space of the rows of $\mathbf{K}$ off S. The constrained least-squares fit of $\mathbf{b} - \mathbf{Ax}$ on the basis $\mathbf{KN}$ gives the covariance of Section 10. $\mathbf{x}$ is not changed. For `enforce_inv`=3 these are the refit's own errors; for 2 those the refit would have. Neither includes the selection or the shrinkage.

Implementation. `lasso_support_errors` (`constrained_fit.f90`), using `null_space_qr` (now public) and `fit_kernel_basis`; file `svd-lasso.dat`; log lines `LASSO_SUPPORT_ERRORS`. Commit `1a13984`.

Validation. $MoTe_2$: sd(1..3) = 2.8×10^{-5}/0.033/142 (`enforce_inv`=0), 3.0×10^{-5}/0.034/62.8 (2), 3.0×10^{-5}/0.029/47.8 (3); for 3 this refit reproduces LASSO's refit to 1.4×10^{-6}; FCs byte-identical to 9.8.

22 The smooth SVD filter as a per-mode penalty; the ridge prior (9.9)

Motivation. The SVD fits (`enforce_inv`=0, 1) damp poorly determined directions with a smooth filter, while the LASSO cut them with a hard threshold. The four settings should use one regularization. A Gaussian prior on the FCs was also requested as an option.

Algorithm. Let $\mathbf{AK} = \sum_k w_k \mathbf{u}_k \mathbf{v}_k^T$ and $w_c = c\, w_{\max}$. `svbksb` returns

$$\mathbf{y} = \sum_k \frac{w_k^2}{\sqrt{w_k^6 + w_c^6}} (\mathbf{u}_k \cdot \mathbf{b})\, \mathbf{v}_k = \sum_k \frac{1}{\sqrt{1 + (w_c/w_k)^6}} \frac{\mathbf{u}_k \cdot \mathbf{b}}{w_k} \mathbf{v}_k. \tag{26}$$

The factor $1/\sqrt{1 + (w_c/w)^6}$ multiplying the plain inverse is the magnitude response of a third-order Butterworth filter in the variable w_c/w. Below the cut the gain is w^2/w_c^3; ridge regression, $w/(w^2 + w_c^2)$, gives w/w_c^2. Minimizing $\|\mathbf{AKy} - \mathbf{b}\|^2 + \sum_k g_k (\mathbf{v}_k \cdot \mathbf{y})^2$ gives $\mathbf{v}_k \cdot \mathbf{y} = w_k (\mathbf{u}_k \cdot \mathbf{b})/(w_k^2 + g_k)$; equating with Eq. (26),

$$g_k = w_k^2 \Big(\sqrt{1 + (w_c/w_k)^6} - 1 \Big). \tag{27}$$

So the filter is a quadratic penalty with a mode-dependent weight, and it can be added to the LASSO objective. In the variables of Section 20, $\mathbf{y} = \mathbf{B}^{-1}\mathbf{t}$ with $\mathbf{B} = \mathbf{Q}^T \mathbf{W} \mathbf{K}_r$, and the penalty $(1/2N) \sum_k g_k y_k^2$ is $\frac{1}{2}\mathbf{t}^T \mathbf{P} \mathbf{t}$, $\mathbf{P} = \mathbf{B}^{-T} \mathrm{diag}(g) \mathbf{B}^{-1}/N$; $\mathbf{K}_r$ holds the right singular directions of $\mathbf{AK}$ with $w_k > 10^{-15} w_{\max}$. Without the L_1 term the fit is that of `svd_set`.

Bayesian reading. With Gaussian noise of variance σ^2 and a Gaussian prior of width τ on each parameter, the posterior maximum minimizes $\|\mathbf{Ax} - \mathbf{b}\|^2/(2\sigma^2) + \|\mathbf{x}\|^2/(2\tau^2)$: ridge with $\lambda_2 = \sigma^2/\tau^2$. The filter is the same with a prior width that depends on the mode, $\tau_k^2 = \sigma^2/g_k$.

Optional ridge prior. With `default.params` line 7 = 1 the LASSO objective gets

$$\frac{s^2}{N} \sum_j \frac{x_j^2}{2\tau_{r(j)}^2}, \tag{28}$$

with s^2 the residual variance of the full filtered least-squares fit and τ_r the rms of the rank-r FCs of that fit. It enters the LASSO, the refits and the cross-validation training fits; the residuals (CV errors, noise test) are those of the forces alone.

Implementation. `classo_cv` (`resolved_kernel` with 10^{-15}), `classo_cv_res` (internal `filter_penalty`; the ridge term with `l2fac`, `colrank`) in `lasso_constrained.f90`; switch read in `ios.f90` (only the value 1 switches it on; the environment variable `FOCEX_L2_TAU` scales τ, for tests only). Commit `7f89a43`. The log/ `lasso.dat` lines give the number of directions below the filter scale and, with line 7, s^2 and τ.

Validation. ZnO cubic exact model, FC3 to 7 shells while the model has 5, 21 600 rows, force noise 10^{-3}: the run with the hard cut equals the run with the filter (no direction below the filter scale in these data). Ridge with $\tau \times 0.5$, 1, 2: the same within one FC (the term is negligible with these data). Not tested: noisier data or fewer snapshots.

23 The noise test for the LASSO weight (9.9)

Motivation. Cross-validation chose $\lambda/\lambda_{\max} = 1.08 \times 10^{-4}$ on LaP (FC3 to 6 shells) and kept terms at the noise level. The weight should follow from the force noise.

Algorithm. Fit all resolved parameters (filtered least squares): residual $\mathrm{RSS}_{\mathrm{full}}$, effective number of parameters p_{full}, and $s^2 = \mathrm{RSS}_{\mathrm{full}}/(N - p_{\mathrm{full}})$. On the all-rows path, for each λ refit on the support and compute $\mathrm{RSS}(\lambda)$ and $p(\lambda)$. The effective number of parameters is the trace of the hat matrix of the refit, $p = \mathrm{Tr}[(\mathbf{N}^T \mathbf{G}_p \mathbf{N})^+ \mathbf{N}^T \mathbf{G} \mathbf{N}]$, with $\mathbf{G}_p$ the Gram matrix including the penalties and $\mathbf{G}$ that of the data. With $\Delta p = p_{\mathrm{full}} - p(\lambda)$, if the dropped parameters were zero, $\Delta\mathrm{RSS}/s^2$ would be $\chi^2_{\Delta p}$, of mean Δp and variance $2\Delta p$. The rule takes the largest λ with

$$\mathrm{RSS}(\lambda) - \mathrm{RSS}_{\mathrm{full}} \le s^2 \big(\Delta p + 2\sqrt{2\Delta p} \big) \tag{29}$$

(mean plus two standard deviations; a round-off allowance $10^{-13}N\langle b^2\rangle$ is added). With exact forces $s = 0$, and only refits that reproduce the full fit pass. If none passes, the grid is extended as for the CV.

Implementation. `default.params` line 6 = 1 (`lasso_rule`); `classo_cv_res`, `classo_path` (argument `npf`), `refit_support` (argument `pf`) in `lasso_constrained.f90`. `lasso.dat` lists, per λ, RSS − $\mathrm{RSS}_{\mathrm{full}}$ and the free parameters, and the line `noise test: RSS_full ... p_full ... s^2 ... first passing k`.

Validation. ZnO cubic exact model as above (FC3 to 7 shells, model 5; 229 true FC3). "LASSO+refit" is `enforce_inv`=3 with the filter penalty:

forces	method	$\lambda/\lambda_{\max}$	FC3 $\neq 0$	true lost	false kept	rms err/σ
exact	LASSO+refit, noise test	–	229	0	0	–
noise 10^{-3}	least squares	–	624	–	395	0.99
noise 10^{-3}	LASSO+refit, noise test	5.4×10^{-5}	374	10	155	0.64
noise 10^{-3}	LASSO+refit, CV	4.2×10^{-5}	390	7	168	0.62

With exact forces FC2 and FC3 are exact (5.7×10^{-13}, 3.3×10^{-10}). With noise, $s = 0.99\times10^{-3}$ (true 10^{-3}); the lost terms are all $\le 3.2\sigma$ ($\le 0.4\,\%$ of the largest FC3), the false ones all $< 1.8\sigma$ (median 0.36). About 9 min and 0.34 GB per LASSO run. On these data the noise test and CV are close; data where they differ were not tested.

24 The SVD cut moved into a gap of the singular values (9.9)

Motivation. The filter of Eq. (26) damps a mode below the cut; it does not remove it. A nearly singular mode not far below the cut, or just above it, is amplified. In the MgO case of Section 27 this gave FC2 of order 10^5 eV/Å^2. Raising the cut just above the small values is not enough. With line 2 = 4×10^{-5} the relative cut $c = 1.6\times10^{-9}$ lies above every small value ($r = 1.6\times10^{-10}$), yet FC2 was still 300–500. The gain of that mode below the cut is, in units of $1/w_{\max}$, $r^2/c^3 = 6.2\times10^6$; in absolute units ($w = 1.87\times10^{-11}$, $w_c = c\,w_{\max} = 1.87\times10^{-10}$, $w_{\max} = 0.1167$) $w^2/w_c^3 = 5.3\times10^7$ per unit of $\mathbf{u}_k\cdot\mathbf{b}$, the same number ($6.2\times10^6/0.1167$).

Algorithm. Let $r_k = w_k/w_{\max}$, sorted ascending, and c the relative cut. A value is dangerous if

$$\left(r < c \text{ and } r^2 > 10\,c^3\right) \quad \text{or} \quad c \le r < 10^3 c. \tag{30}$$

Below the cut the gain is $\approx r^2/c^3$; the factor 10 compares it with the gain $1/r$ of a resolved mode at $r \approx 0.1$. Just above the cut the gain is $\approx 1/r$. If there is a dangerous value, let r_{lo} be the largest one; search upward from it for the first gap $r_{k+1} \ge 10^3 r_k$, and set

$$c \leftarrow r_{\mathrm{lo}}^{0.1}\, r_{\mathrm{hi}}^{0.9}, \tag{31}$$

with $r_{\mathrm{lo}} = r_k$, $r_{\mathrm{hi}} = r_{k+1}$ the edges of the gap (nearer the good values). Then the top bad mode has gain $(r_{\mathrm{lo}}/r_{\mathrm{hi}})^{1.7}/r_{\mathrm{hi}}$ and the smallest good mode is changed by $\frac{1}{2}(r_{\mathrm{lo}}/r_{\mathrm{hi}})^{0.6}$, at most 8×10^{-3} for a gap of 10^3 (code comment). Without such a gap the cut is left unchanged and a message is printed. The cut is never lowered.

Implementation. `gap_cut` (`svd_ridge.f90`), called by `svd_set` and by `classo_cv_res` for the LASSO filter. Screen and log: `SVD: singular values next to the cut ...; cut moved to the gap at ...` (or `no gap above them: cut unchanged`). The minimum gap was 10^4 and the new cut the geometric mean of the edges (capped at 10^{-3}) in earlier versions of the same day.

Validation. MgO with Born charges (`fc2range`=0, `enforce_inv`=1, 2 single-atom snapshots): small singular values up to 1.6×10^{-10} of $w_{\max}$, the next 9.1×10^{-2}; cut $1.6 \times 10^{-9} \to 1.21 \times 10^{-2}$ (3.82×10^{-6} with the geometric mean; the FCs agree to 4–5 digits); FC2 group 9 from 3.85×10^{5} to $\leq 5 \times 10^{-3}\,\mathrm{eV/Å^2}$; fit 0.933 % unchanged; largest FC2 11.66. The same with line 2 $= 4 \times 10^{-5}$. ZnO runs without small values: byte-identical apart from timing.

25 Refit when the measured noise exceeds the guess (9.9)

Motivation. The cut $c = \epsilon_F^2$ comes from the user's guess of the force noise (line 2). If the forces (or the model) are much noisier, modes below the noise are fitted.

Algorithm. After the fit, the measured relative noise is

$$\varepsilon^2 = \frac{\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2}{\|\mathbf{b}\|^2}. \tag{32}$$

If $\varepsilon^2 > 100\,c$ (the measured noise more than 10 times the guess) and ε^2 exceeds the current cut, the cut is set to ε^2, passed through `gap_cut` (giving w_c'), and the solution is redone from the same SVD. A large misfit can also be model error (missing FCs), not noise; then a larger cut only removes information. So the refit is kept only if it passes the noise test of Section 23. With the filter factors of `svbksb`,

$$\phi_k = \frac{w_k^3}{\sqrt{w_k^6 + w_c^6}}, \qquad p = \sum_k \phi_k, \tag{33}$$

(p is the trace of the hat matrix, the effective number of fitted parameters), $s^2 = \mathrm{RSS_{old}}/(N-p)$ and $\Delta p = p(w_c) - p(w_c')$, the refit is kept if

$$\mathrm{RSS_{new}} - \mathrm{RSS_{old}} \leq s^2\left(\Delta p + 2\sqrt{2\Delta p}\right) \tag{34}$$

(plus a round-off allowance $10^{-13}\|\mathbf{b}\|^2$); otherwise the first solution and its errors are kept. The cut is only raised.

Implementation. `svd_set` (`svd_ridge.f90`; filter factors in its internal function `phif`), active only when the flag `svd_noise_check` (`common/params.f90`) is set. `focex.f90` sets it around the force fits of `enforce_inv`=0 (`svd_set`) and 1 (`fit_kernel_basis`). It is off for `lasso_support_errors`: there the fitted vector is a residual, whose relative misfit is about 1 by construction (running the check there was a bug, found and fixed on 6 November 2026). The LASSO filter applies the same rule to its full filtered fit (`classo_cv_res`, $\varepsilon^2 = \mathrm{RSS}/\|\mathbf{b}\|^2$). Log lines: `SVD: measured relative noise ... (line 2 gave ...): refit with the cut ...`, then `SVD: refit kept (noise test passed): parameters dropped ..., RSS increase relative ...` or `SVD: refit rejected: ... model error; cut unchanged`.

Validation. Runs of the final 9.9 build:

- MgO with Born charges (2 single-atom snapshots, default FC2 range, FC3 3 shells, `enforce_inv`=1). With the default line 2 ($c = 10^{-10}$) the cut moved into the gap (Section 24), and the measured noise $\varepsilon = 1.76 \times 10^{-3}$ triggered a refit, kept by the test (0.99 parameters dropped, RSS increase 1.73×10^{-5} relative). The result equals that of the hand-tuned line 2 $= 4 \times 10^{-4}$: fit 0.174 % / 0.176 %, Γ T_{1u} 534.89 $\mathrm{cm^{-1}}$, highest frequency 748.9 $\mathrm{cm^{-1}}$, no unstable k, Grüneisen median (IBZ, $\omega > 20\,\mathrm{cm^{-1}}$) 1.73 / 1.68.

- $MoTe_2$, FC2 only (the misfit is model error, not noise): first-fit ε from 0.44 to 0.032 for 2 to 10 FC2 shells; every refit was rejected (10 shells: dropping 8.1 parameters would raise the RSS by a factor 324). Without the test the refit had taken the misfit for noise and given a 97.8 % residual.
- ZnO cubic exact model: with exact forces no refit, `fc3.dat` identical; with 10^{-3} noise, refit to the cut 3.0×10^{-6}, FC errors unchanged (FC3 rms $0.99\,\sigma$).

26 Rotational invariants of FC3 and FC4 (9.9)

Motivation. The range of FC3 and FC4 is to be judged from their size against distance, in a form that does not depend on the axes.

Algorithm. For each triplet (quadruplet) of atoms, the Frobenius norm $\|\Psi\| = (\sum_{\alpha\beta\gamma} \Psi^2_{\alpha\beta\gamma})^{1/2}$ (likewise $\|\chi\|$), and for FC4 the isotropic part

$$\mathrm{iso} = \frac{1}{3} \sum_{\alpha\beta} \left(\chi_{\alpha\alpha\beta\beta} + \chi_{\alpha\beta\alpha\beta} + \chi_{\alpha\beta\beta\alpha} \right), \tag{35}$$

both invariant under rotations, against the largest pair distance $d_{\max}$ of the tuple and the number of distinct atoms. Tuples that differ only by the order of the last indices are written once.

Implementation. `write_fc_invariants` (`ios.f90`): `trace_fc.dat` blocks 1 (FC3) and 2 (FC4); plotted by `FOCEX/PLOTTING/fcs.plt`. Commit `1a13984`.

Validation. $MoTe_2$: 506 triplets, norms equal to a Python recomputation from `fc3.dat` to 3×10^{-5} (rounding). The FC4 plot branch was tested for gnuplot syntax only.

27 Wigner–Seitz boundary aliasing

Problem. In a supercell with lattice $\{L\}$, a displacement of atom j moves all its images. The force on atom i sees $\sum_L \Phi(i, j+L)$. If a pair (i, j) has several images at the same distance on the boundary of the supercell Wigner–Seitz cell, the data fix only their sum. With d_k the image vectors related by point operations R_k ($\Phi(d_k) = R_k \Phi(d_1) R_k^T$), the data see

$$\Phi(d_1) \;\mapsto\; \sum_k R_k \, \Phi(d_1) \, R_k^T, \tag{36}$$

restricted to the components that the site symmetry of d_1 allows. Combinations in the kernel of this map are invisible to any data of this supercell; equal weights of the images do not change the rank.

MgO case. Run with Born charges, `fc2range`=0 (whole supercell), `enforce_inv`=1, 2 single-atom snapshots of 0.0106 Å. FC2 group 9 (Mg–O at 6.991 Å) came out at $\pm 10^5$ eV/Å^2 (385136, -208352, …), sd(2) 9.7×10^4, Hill B printed as -4×10^7. The pair lies on the boundary of the 4×4×4 rhombohedral supercell: three images $v_1 = (-\frac{3}{2}, -\frac{1}{2}, -\frac{1}{2})$, $v_2 = (\frac{1}{2}, -\frac{1}{2}, \frac{3}{2})$, $v_3 = (\frac{1}{2}, \frac{3}{2}, -\frac{1}{2})$ (units of the cubic a), all at $a\sqrt{5.5} = 6.991$ Å. The site group of v_1 has order 2, leaving 5 components. The image-sum map has singular values 3, 3, 0, 0, 0: rank 2 of 5. Three combinations are invisible. Their singular values in `svd-kernel.dat` were round-off (5×10^{-17} to 1.5×10^{-14}, with $w_c = 1.17 \times 10^{-11}$). The filter gain w^2/w_c^3 (absolute, per unit of $\mathbf{u}_k \cdot \mathbf{b}$) is 1.3×10^5 at $w = 1.45 \times 10^{-14}$ (ridge would give $w/w_c^2 \approx 10^8$, a hard cut 0); hence FC2 $\pm 10^5$. The same gain arises in the LASSO filter. The earlier runs with `fc2range`=1 (6 shells) were fine: largest FC2 11.66, fit 2.02 % (Born), 7.20 % (no Born).

Remedy and open points. The gap rule of Section 24 moves the cut above these modes and gives FC2 of group $9 \leq 5 \times 10^{-3}\,\mathrm{eV/Å^2}$. Proposed, not done: zero modes below a round-off level (about $10^{-13} w_{\max}$), or keep only the visible combination of boundary FC2 groups (or drop rank-deficient boundary groups). Not checked in the code: why the keep rule kept group 9. A related case was recorded in 9.1: on a ZnO 3×3×2 supercell, groups at 5.0–6.0 Å lie wholly on the boundary, only $\Phi(d) + \Phi(d + L)$ is fixed, the FCs are off by 0.1 and the bands by $11.8\,\mathrm{cm^{-1}}$, exact only at the commensurate k; a 4×3×2 supercell with the same shortest vector gives every FC2 exactly.

28 Invariances with an inconsistent model: $MoTe_2$

Motivation. FOCEX never gave good results for 2H-$MoTe_2$: unstable interlayer modes and negative elastic constants. $MoTe_2$ is a layered crystal whose interlayer forces are weak. A small inconsistency between the data, the truncated model and an imposed invariance then lands in the soft interlayer FCs. This section separates the causes. System: 6 atoms per cell, 4×4×2 supercell of 192 atoms. `FORCEDISP1`: the reference and 6 single-atom snapshots; `FORCEDISP2`: 33 single-atom and 376 pair snapshots, pair distances 2.705–9.257 Å. All runs with the 9.9 build on 6 November 2026.

28.1 Data consistency check

Algorithm. Forces are short-ranged after subtraction of the reference forces. For each snapshot s with displaced atoms D_s take the far atoms $\mathcal{F}_s = \{i : \min_{j \in D_s} |X_i - X_j| > R_{\mathrm{far}}\}$ and compute

$$\delta_s = \Big[\frac{1}{3|\mathcal{F}_s|} \sum_{i \in \mathcal{F}_s} \big|F_i^{(s)} - F_i^{\mathrm{ref}}\big|^2\Big]^{1/2}. \tag{37}$$

In a consistent data set δ_s is at the noise level. A snapshot with a large δ_s was computed with a different reference (other settings, unconverged, other structure).

Result. $R_{\mathrm{far}} = 9$ Å. `FORCEDISP1` snapshots 2–4 (displacement 0.0100 Å) give $\delta = 1.7 \times 10^{-3}\,\mathrm{eV/Å}$, with far forces about 0, while the reference has Te forces $\pm 4.906 \times 10^{-3}\,\mathrm{eV/Å}$ along z. Snapshots 5–7, and 402 of the 409 `FORCEDISP2` snapshots, agree to $< 2 \times 10^{-4}\,\mathrm{eV/Å}$ (median 4×10^{-5}). Without snapshots 2–4 the FC2-only fit with 10 shells goes from 12.7 % to 3.2 %. This check is recommended as a data diagnostic; it is not part of FOCEX.

28.2 Huang rows with zero stress at a reference out of equilibrium

Without `stress.params` the Huang rows impose Eq. (7) with $\sigma = 0$, but this reference is not at equilibrium (Te forces above). FC2 to 10 shells, `enforce_inv`=1: interlayer shear mode $E^2_{2g} = -64.8\,\mathrm{cm^{-1}}$, 93 unstable k-points on the path, fit 3.23 %. With `ihuang`=0: $E^2_{2g} = +26.3\,\mathrm{cm^{-1}}$, none unstable, fit 1.63 %. `enforce_inv`=0 (Huang rows only fitted): $+27.6\,\mathrm{cm^{-1}}$.

28.3 FC2 range

With `ihuang`=0 and the single-atom data:

FC2 shells	4	6	8	10	12	14
fit (%)	5.65	3.04	2.35	1.63	1.60	1.55
E^2_{2g} ($\mathrm{cm^{-1}}$)	39.0	22.4	25.3	26.3	25.5	25.3

10 shells were chosen. The elastic constants are then (GPa) C_{11} 150.3, C_{12} 92.9, C_{13} 15.6, C_{33} 46.4, C_{44} 25.0, C_{66} 28.7. The default FC2 range (`fc2range`=0) stops in `force_constants` (atom not found): it needs a larger neighbour search (`default.params` line 3).

28.4 Rotational invariance with truncated FC3

All 413 consistent snapshots, FC2 to 10 shells, `ihuang`=0, `enforce_inv`=1:

FC3 shells, `irot`	fit (%)	B_{1g} (cm^{-1})	unstable k	C_{33} (GPa)	Grüneisen
2, `irot`=1	2.79	−29.2	74	−30.3	−7.9..0.7, median −0.09
3, `irot`=1	1.24	28.9	0	28.2	–
2 or 3, `irot`=0	0.84	37.3	0	45.7	0.0..1.08, median 0.60

B_{1g} is the breathing mode. `enforce_inv`=0 with 2 FC3 shells and `irot`=1 gives the same as `enforce_inv`=1 (B_{1g} −29.1 cm^{-1}, C_{33} −30.1 GPa). With `irot`=0 the results for 2 and 3 FC3 shells are identical: E^2_{2g} 26.3 cm^{-1}; C_{11} 118.3, C_{12} 18.0, C_{13} 13.9, C_{33} 45.7, C_{44} 26.5, C_{66} 50.2 ($= (C_{11} - C_{12})/2$) GPa; long-wave C_{11} 118.4, C_{33} 45.7, C_{44} 23.1 GPa. FC3: 102 independent, 97 of them above 2σ.

Interpretation (not proven). The rank-3 rotational relations of Section 2 tie FC2 to FC3. With FC3 truncated far shorter than FC2 (2–3 shells against 10) they cannot hold for the true FCs, and imposing them distorts the soft interlayer FC2.

28.5 Relaxation

`default.params` line 5 = 1 on the `irot`=0 fit (no `stress.params`: the reference stress from FC1): Te moved by $\pm 2.257 \times 10^{-4}$ Å along z, consistent with force over stiffness ($4.906 \times 10^{-3}/2.257 \times 10^{-4} = 21.7\,\mathrm{eV/Å^2}$); strain -1.19×10^{-4} in plane and $+1.58 \times 10^{-4}$ along c; largest stress after relaxation 1.4×10^{-12} GPa; 97 s. The Γ modes change by less than 0.05 cm^{-1}; the C_{ij} are unchanged.

Lessons. For $MoTe_2$ the problems came from the inputs and the constraints, not from the fitting algorithm: three inconsistent snapshots, Huang rows with $\sigma = 0$ at a reference out of equilibrium, and rotational rows of rank 3 with FC3 truncated short. Check the data first (far-atom forces); give the reference stress, or switch the Huang rows off; switch the rotational rows off when FC3 is much shorter than FC2.

29 Conclusions

Versions 9.0–9.9 changed three things in FOCEX. First, the constraints: invariance rows of all ranks are built in one loop, the Huang rows hold for a stressed reference, the structure and tensors are symmetrized, and the Coulomb part of `born_flag`=2 enters the rows consistently. Second, the physics after the fit: elastic constants by Born–Huang and by long waves, B_0' from FC3, a quasi-harmonic model with B_T and $d \ln L/dT$, Grüneisen parameters with an exact Γ limit, a relaxation that is exact on the test models (with point charges through the full Ewald energy), and symmetry labels of the Γ modes. Third, the fit itself: standard errors, a constrained LASSO with exact invariances, and in 9.9 one regularization for all settings (the smooth filter as a penalty), a noise-based choice of λ, and rules that protect the fit against nearly singular modes and against misfit that is model error. The $MoTe_2$ study shows that, for a soft layered crystal, inconsistent data and invariances imposed on a truncated or out-of-equilibrium model matter more than the fitting method. Table 1 lists the changes.

Open points recorded in the sources.

- The data consistency check of Section 28 is not part of FOCEX.
- $MoTe_2$: the explanation of the rank-3 rotational rows is an interpretation, not proven.

Table 1: Summary of the changes. "Switch" is the input that turns a change on, if any.

Version	Change	Switch
9.0	Invariance rows of all ranks in one loop; rotational offsets fixed	–
9.0	Z^*, ϵ checked against the space group	`born_flag`$\neq 0$
(9.1 tag)[a]	Range of FC3–FC8 clusters	`structure.params` l. 12 flags
(9.1 tag)[a]	Generalized Huang rows, stressed reference	`stress.params`; Huang flag
(9.1 tag)[a]	Constrained LASSO, exact invariances, refit	`enforce_inv`=2, 3
9.1	Frequencies at the exact k; Γ as a limit	–
9.2	Born–Huang elastic constants; B_0' from FC3; QHA with B_T, $d\ln L/dT$	–
9.2	Mode Grüneisen parameters at k, long-wave limit at Γ	–
9.2	Standard errors sd(rank)	–
9.3	Relaxation of an unrelaxed reference	`default.params` l. 5 = 1
9.4	IR/Raman activity, irreps, computed character tables; `tol_sym`; Γ threshold	–
9.5	Some ranks read, others fitted	`include_fc`=2
9.5	Hybrid Coulomb subtraction, Coulomb right-hand sides	`born_flag`=2
9.5	Elastic constants from long-wave sound speeds	–
9.5	Irreducible wedge from the Laue group (shared)	–
9.6	Symmetrized positions, Z^*, ϵ, stress	–
9.7	OpenMP SVDs, `-O3`, identical results	`FOCEX_NUM_THREADS`
9.7	Coulomb FCs (second order) in the relaxation	`born_flag`=2, l. 5 = 1
9.8	Exact Coulomb relaxation by Broyden; relax-only reruns	$Z^* = z\!\!\!/$; all `include_fc`=2
9.9	Standard errors for LASSO fits	`enforce_inv`=2, 3
9.9	SVD filter as per-mode penalty in the LASSO	`enforce_inv`=2, 3
9.9	Noise test for λ	`default.params` l. 6 = 1
9.9	Ridge prior, τ = rms per rank	`default.params` l. 7 = 1
9.9	SVD cut moved into a singular-value gap	–
9.9	Refit when measured noise $> 100\times$(l. 2)^2, kept if it passes the noise test	– (force fits)
9.9	FC3/FC4 invariants in `trace_fc.dat`	–

[a] Commit `c9edf0a`, 27 October 2026: after the 9.0 tag, inside the 9.1 tag; not in the manual's history.

- Noise test and ridge prior: tested only on one exact model with 10^{-3} noise; noisier data or fewer snapshots not tested.
- Wigner–Seitz boundary groups: the gap rule treats the symptom; removing the invisible combinations is proposed, not done.
- Anisotropic Z^*: the relaxation keeps the second-order Coulomb route; the rank-4 rotational right-hand side for tensor ϵ is not implemented.
- The FC3 standard errors are 10–15 % larger than the observed errors on the exact model; cause unknown.
- Rotational rows raise the held-out force error in cross-validation on GaN and ZnO (recorded at 9.6); cause open.
- ZnO long-wave C_{33} 45–49 GPa in every setting, against about 210 measured (recorded at 9.6).